

BASIC SQL INTERVIEW QUESTIONS

1. What is SQL?
 2. What are the different types of SQL commands (DDL, DML, DCL, TCL)?
 3. What is the difference between DELETE, TRUNCATE, and DROP?
 4. What is a primary key? Can a table have multiple primary keys?
 5. What is a foreign key? Why is it used?
 6. What is the difference between WHERE and HAVING?
 7. What is the difference between CHAR and VARCHAR?
 8. What is NULL in SQL? How is it different from 0 or empty string?
 9. What are aggregate functions? Name a few.
 10. What is the use of DISTINCT keyword?
 11. What is the difference between COUNT(*) and COUNT(column_name)?
 12. What is the default sort order of ORDER BY?
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INTERMEDIATE SQL INTERVIEW QUESTIONS

13. What is a JOIN? Explain different types of JOINS.
 14. Difference between INNER JOIN and LEFT JOIN with example.
 15. What is a self join? Where is it used?
 16. What is a subquery? Types of subqueries.
 17. What is the difference between a subquery and a JOIN?
 18. What are indexes? Why do we use them?
 19. Difference between clustered and non-clustered index.
 20. What is a composite index?
 21. What is normalization? Explain 1NF, 2NF, 3NF.
 22. What is denormalization? When do we use it?
 23. What is a view? Can we update data using a view?
 24. Difference between UNION and UNION ALL.
 25. What is GROUP BY? Can we use GROUP BY without aggregate functions?
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ADVANCED SQL INTERVIEW QUESTIONS

26. What is a stored procedure? How is it different from a function?

27. What are triggers? Types of triggers in MySQL.
 28. What is a transaction? Explain ACID properties.
 29. What is deadlock? How can it be handled?
 30. What is the difference between RANK(), DENSE_RANK(), and ROW_NUMBER()?
 31. What are window functions? Give real-world use cases.
 32. What is query optimization? How do you optimize slow queries?
 33. What is execution plan? How do you analyze it in MySQL?
 34. What are common causes of poor SQL performance?
 35. What is EXISTS vs IN? Which performs better and why?
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SCENARIO-BASED SQL INTERVIEW QUESTIONS (Very Important)

36. Employee Table Scenario

You have an employees table.

Find the **second highest salary** without using LIMIT.

37. Duplicate Data Scenario

A table has duplicate email IDs.

How will you identify and remove duplicates?

38. Sales Analysis Scenario

You have orders(order_id, user_id, amount, order_date).

Find total sales **month-wise**.

39. Data Integrity Scenario

Users are getting deleted but orders must stay.

How will you design foreign key constraints?

40. Performance Scenario

A query works fast in development but slow in production.

What steps will you take to debug?

41. Login System Scenario

You need to track failed login attempts.

How will you design the table & query?

42. Analytics Scenario

You want **top 3 products per category**.

How will you write the SQL?

43. Concurrency Scenario

Two users update the same record simultaneously.

How does SQL ensure consistency?

44. Large Data Scenario

A table has 50 million records.

How do you improve report query performance?

45. SQL query feeds a dashboard and runs every 5 minutes.

What SQL-level optimizations will you apply?